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Marks	1.00/1.00
Grade	**3.75** out of 3.75 (**100**%)

Question 1

Correct

Mark 1.00 out of 1.00

What is the relative position of these two lines in the plane?

$-4 \cdot x + 5 \cdot y = 0$
$20 \cdot x - 25 \cdot y = 0$

a. (none of the above)
b. Coincident ✓
c. Parallel
d. Non perpendicular secants
e. Perpendiculars

Given the two straight lines, we consider the system of two equations and two variables that they determine.

In this case, the position of the two straight lines can be:

1. coincident (when the system is consistent and dependent - the two straight lines have infinitely many points in common and therefore they have to be coincident)

2. parallels (when the system is inconsistent - the two straight lines do not have any point in common)

3. secant or intersecting (when the system is consistent and independent - they intersect at a unique point)

In the case of intersecting lines, additionally, the two lines will be perpendicular if their director vectors are two orthogonal vectors. Recall that in a straight line given in the form

Ax+By+C=0

the director vector is v= (-B,A).

In this case, the augmented matrix of the system of two equations and two variables determined by the two straight lines is:

$$\begin{pmatrix} -4 & 5 & 0 \\ 20 & -25 & 0 \end{pmatrix}$$

To find out the kind of system of equations we can use Gaussian elimination yielding to

$$\begin{pmatrix} -4 & 5 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

where the rank of the matrix is rank(M) = 1 and the rank of the augmented matrix is rank(A) = 1.

Case 1: $rank(M) = rank(A) = 1 \rightarrow$ coincident straight lines

Case 2: $rank(M) = 1 \ and \ rank(A) = 2 \rightarrow$ parallel straight lines

Case 3: $rank(M) = rank(A) = 2 \rightarrow$ secant straight lines

Additionally, if the straight lines are secant, they will be perpendicular if their director vectors are two orthogonal vectors. In this case the two director vectors are $(-5,-4)$ and $(25,20)$, and therefore its scalar product is

$(-5,-4) \cdot (25,20) = -205$.

Therefore, the relative position of the two straight lines is: Coincident.

The correct answer is: Coincident